

Haocheng He

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PROFESSIONAL SUMMARY

Industrial Engineering graduate from Guangdong University of Technology, with a minor bachelor's degree in Law, and currently pursuing an MSc in Aviation Engineering and Operations Management at The Hong Kong Polytechnic University. My experience spans four internships in AI hardware integration, airport emergency simulation, manufacturing improvement, and assembly-line optimization, along with a time-series RUL prediction project. I learn quickly, enjoy digging into technical problems, and am comfortable working across hands-on engineering, data modeling, and high-pressure execution.

EDUCATION

The Hong Kong Polytechnic University Hong Kong SAR, China
Master of Science in Aviation Engineering and Operations Management Sep 2025 – Jan 2027
GPA: 3.95 / 4.3 (Current)
Thesis: Simulation Modelling and Optimization for Airport Ferry Vehicle Scheduling under Uncertainty

Guangdong University of Technology Guangzhou, China
Bachelor of Engineering in Industrial Engineering Sep 2021 – Jun 2025
GPA: 3.67 (Top 20%)
Thesis: AnyLogic-Based Simulation Optimization of Emergency Vehicle Route Planning in Large Airports
Honors: Excellent Graduation Thesis + Innovation Award

Bachelor of Laws (Minor Degree in Law) Sep 2022 – Jun 2025
GPA: 3.61
Thesis: Coordination Between Copyright Protection and Privacy Protection in Digital Rights Management

EXPERIENCE

AI Engineer Assistant May 2025 – Jun 2026
Shenzhen Wuxian Hongyin Technology Co., Ltd. Shenzhen, China

- Supported module integration for an Ascend-based piano fingering recognition system (Python/C++); reviewed architecture and helped locate issues.
- Maintained Ubuntu servers and Huawei switches/routers; also supported procurement, administration, and technical interviews.

Research Assistant Sep 2023 – Jun 2025
Guangdong University of Technology (with Guangzhou Baiyun Int'l Airport) Guangzhou, China

- Built an airport road-network, apron, and runway-area simulation model in AnyLogic, Java, and MySQL for aircraft landing and emergency fire-truck dispatch scenarios.
- Designed single/multi-vehicle, single/multi-event, and runway-road-closure scenarios; compared D* Lite, A*, Dijkstra, BFS, and GA routing algorithms.
- Completed 4 emergency-scenario comparisons across 5 routing algorithms; D* Lite achieved the highest average reliability index (1.342).

Industrial Engineering Intern Apr 2024 – May 2024
Midea Group — Kitchen & Water Heater Division Foshan, China

- Conducted stopwatch time studies across all workstations on a dishwasher sheet-metal line, built operator workload tables, and identified 5 over-cycle steps.
- Proposed 22 improvements using lean methods; simulation showed line balance improving from 58% to 70%, reducing headcount from 20 to 18.

- Timed 12 workstations on the wheel loader assembly line with a stopwatch; identified 3 major bottlenecks using ECRS analysis.
- Proposed layout adjustments estimated to reduce cycle time by approximately 8% in simulation. Submitted recommendations to the production-line team for evaluation.

PROJECTS

Aircraft Engine Remaining Useful Life Prediction (Course Project)

Python / C-MAPSS

- Built an RUL prediction workflow on NASA C-MAPSS FD001 multivariate time-series data; cleaned operating-condition data and 21 sensor signals, constructed sequence features, and applied PCA.
- Compared LSTM, BiLSTM, Transformer, and Random Forest baselines for time-series forecasting; after iterative tuning, the BiLSTM model reached about 8% final error from an initial model error of about 50%.

SKILLS

Programming & Data: Python, Java, SQL/MySQL, C/STM32, JavaScript/TypeScript

AI & Automation: Claude Code, Codex, Hermes, ChatGPT, Prompt Engineering, Loop Engineering (Agent Loop Design), AI Agent workflows, model deployment/evaluation

Machine Learning: Random Forest, LSTM, BiLSTM, Transformer, PCA, model training/evaluation

Simulation & Optimization: AnyLogic, Plant Simulation, OR-Tools, Dijkstra, A*, BFS, genetic algorithms, line balancing, ECRS

Engineering Tools: Ubuntu/Linux, Git, LCEDA, oscilloscope, logic analyzer, Huawei switches/routers

Languages: English (IELTS 6.5), Mandarin (native), Cantonese (conversational)

INTELLECTUAL PROPERTY

Published Invention Patent Applications

- *MCMC Sampling-Based Motion Control Method for a PI Coating Machine Controller*. CN122085794A, Guangdong University of Technology, 2026. Inventors: Xianzhong Zhou, Haocheng He, Zhiyuan He.
- *Coating Control Method and System Based on Deep Reinforcement Learning*. CN121325552A, Guangdong University of Technology, 2025. Inventors: Xianzhong Zhou, Haocheng He, Zhiyuan He.
- *Energy Regulation Method and System for a Household Solar Energy Storage System*. CN117767437A, Guangdong University of Technology, 2024. Inventors: Xianzhong Zhou, Huangtao Zeng, Daming Yu, Haocheng He.

Utility Model Patent

- *Integrated Equipment for Wire Harness Production*. CN221352455U, Guangdong University of Technology, 2024. Inventors: Xiang Zhao, Haocheng He, Bofeng Hu.

Computer Software Copyright Registrations

- Enterprise Financial Data Analysis Software V1.0. Reg. No. 2024SR0146827, 2024. Authors: Yisi Li, Haocheng He.
- Urban Flood Intelligent Mobile Emergency Command & Dispatch System V1.0. Reg. No. 2024SR0549247, 2024. Authors: Chengfeng Lin, Zeyin Li, Hua Yang, Haocheng He.